READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27-40**, which are based on Reading Passage 3 on pages 10 and 11.*

Animals predicting earthquakes?

Surely it is too much to believe that animals can predict earthquakes when we haven't worked out how to do it ourselves?

On 26 December 2004, villagers from Bang Koey in Thailand noticed buffalo on the beach lift their heads, look out to sea, then stampede to the top of a nearby hill. Minutes later, the tsunami* struck. Could these creatures have been sensing early warning signs of the earthquake that triggered the Asian tsunami? It is a strange assertion but the possibility that animals might hold the answer cannot be ignored, as a warning system could save many lives.

The idea that animals can predict earthquakes has ancient origins. From 373 BC there are stories of rats, dogs and snakes deserting the Greek city of Helice before an earthquake hit. It was the first in a long line of such anecdotes. What has been lacking is any real scientific data linking animal behaviour with earthquakes.

In late 2000, however, Stanley Coren from the University of British Columbia started a study to discover whether dogs suffer from 'seasonal affective disorder.'** Twice weekly, he emailed 200 dog owners in Vancouver, asking them to rate their pets' activity and anxiety levels using a nine-point scale. In general there was little daily variability; Coren's initial analysis of many months' worth of information strongly refuted his suspicion that dogs become depressed during winter, and he abandoned the project. When he finally went through the figures in detail several years later however, he noticed that on 27 February 2001, nearly 50 per cent were well above their usual baseline for activity and anxiety. The likelihood of such a difference was remote and Coren searched through the newspaper archives for a cause. He found that on 28 February a quake of magnitude 6.8 shook the Pacific north- west, with an epicentre at Nisqually about 240 kilometres south of Vancouver. Coren wondered what the dogs could have sensed. There was little to guide him.

Even the most comprehensive earthquake research project in the world, the Parkfield experiment in California, couldn't help. Since 1985, researchers from the US Geological Survey (USGS) have monitored a section of the San Andreas fault near the town of Parkfield. They have analysed every tremor, and measured tiny movements of the fault at 10- minute intervals, but to date the project has revealed nothing that would reliably indicate that an earthquake is imminent.

There is still much speculation about animals predicting earthquakes. For example, one idea is that some animals detect changes in the Earth's electrical field. Another theory is that animals are responding to gas such as radon released from rocks before a quake, despite the fact that few experts accept such gas is produced. Rupert Sheldrake, a British researcher, suggests that 'people notice unusual animal behaviour before disasters. I think animals are picking up on something that we can't, perhaps by using a sixth sense.'

Coren, however, had read about rescue dogs hearing avalanche victims under the snow, and suspected the dogs in his study might also be hearing vibrations. So he returned to his data to search for supporting evidence. He discovered that only one of the 14 hearing- impaired dogs in his study had shown any significant increase in anxiety on 27 February, and it was living with a hearing dog that had become anxious. Dogs with floppy ears showed only half the change in activity and one-third the change in anxiety level of dogs with pricked ears. Not only would an ear flap reduce the amount of sound reaching the inner ear; Coren also realised that it would weaken high frequency sounds more than low frequency ones. Further, dogs with smaller heads were significantly more likely to behave strangely than those with larger heads, which was interesting, given that smaller-headed dogs are more sensitive to higher frequencies.

Taken together, Coren's results present an alluring hypothesis. He suggests that the kind of high frequency sounds that many dogs can hear are emitted before an earthquake, perhaps from rocks scraping underground. Admittedly this is only one study. Even if Coren is right, is it still possible that other animals could predict quakes in different ways? Eric Wikramanayake, a conservation scientist who used radio collars and happened to be studying elephants in Sri Lanka when the tsunami struck, is entirely sceptical.

What he found was precisely nothing. One herd was only 100 metres away from the beach when the tsunami arrived and they just took cover behind a large sand dune when the wave came in sight. The other herd was safely located about 5 kilometres inland and did not show any unusual movements.

Even if some quakes are preceded by high- frequency vibrations, is it feasible that dogs in Vancouver could detect sounds emitted near Nisqually? USGS seismologist Andy Michael points out that the epicentre was over 240 kilometres south of Vancouver. 'It is physically implausible for seismic waves in the kilohertz range to travel that far,' he says. Although unconvinced by Coren's ideas, Michael adds a cautionary tale. 'When Alfred Wegener presented his theory of continental drift, his core ideas about tectonics were right but nobody listened because the mechanisms were wrong.'

So if animals are able to foretell earthquakes, does it matter how? In fact, while western society has been reluctant to use animals as earthquake predictors, China has already embraced the idea. Experts there use a video link to keep a 24-hour watch on snakes in farms nationwide. If they try to escape from their enclosure, observers raise a warning.

Other governments may not be willing to go that far just yet, but Coren believes that a centre that people could call when they see unusual things would be useful. 'We would be able to collect much more data and give warnings when we get hundreds of calls from a single area,' he says. It could cost little more than the price of setting up a phone line. Put that way, what have we got to lose?

* tsunami: large, destructive wave caused by earthquakes

** seasonal affective disorder, emotional and psychological changes created by changes in the amount of daylight in winter

Questions 27-30

Do the following statements agree with the views of the writer in Reading Passage 3?

In boxes 27-30 on your answer sheet, write

YES *if the statement agrees with the views of the writer*

NO *if the statement contradicts the views of the writer*

NOT GIVEN *if it is impossible to say what the writer thinks about this*

27 There is historical scientific evidence to support stories of animals predicting earthquakes.

28 Coren's decision to investigate the possibility that dogs can anticipate an earthquake was based on some archive information.

29 Dogs may be more sensitive to changes in the Earth's electrical field than other animals.

30 Coren's findings from his original study of dogs clearly prove they can hear high ofrequencies which signal earthquakes.

Questions 31-35

Write the correct letter in boxes 31-35 on your answer sheet.

*Choose the correct letter, **A**, B, **C** or **D**.*

31 What is the writer's main point in paragraph 4?

A The strength of the next earthquake along the San Andreas fault is unpredictable.

B It is not the first time that the theory of animal prediction has been investigated.

C Experts have never identified any reliable predictors for earthquakes.

D The Parkfield project depended on unreliable methods for earthquake detection.

32 When Coren re-analysed his data for 27 February he found that on that day

A dogs with either poor or good hearing were equally anxious.

B the shape of a dog's ear seemed to correlate with its behaviour.

C dogs with larger heads were the most sensitive to tremors

D the level of activity amongst dogs reduced as the earthquake occurred.

33 Seismologist Andy Michael refers to Alfred Wegener in order to

A suggest there may be an aspect of Coren's theory which is correct.

B dismiss the idea that dogs are capable of sensing earthquakes.

C demonstrate how data can be falsified by subjective researchers.

D criticise certain scientists for not being careful in their research.

34 What is the writer doing in the penultimate paragraph?

A explaining why western researchers do not trust animal behaviour

B arguing how animal prediction is superior to science-based prediction

C suggesting a more reliable animal than the dog for predictive purposes

D illustrating how animal behaviour is already exploited by some researchers

35 What point is the writer making by using the phrase 'what have we got to lose' in the final paragraph?

A The cost of setting up a data collection centre needs to be considered.

B It is worth setting up a call centre to record animal behaviour.

C The pros and cons of different data collection methods must be weighed up.

D There is a danger that existing data may be lost if not stored carefully.

Questions 36-40

Complete each sentence with the correct ending, ***A-H****, below.*

Write the correct letter, ***A-H****, in boxes 36-40 on your answer sheet.*

36 The belief in the ability of animals to predict earthquakes

37 Experts think it is unlikely that gas

38 Sheldrake says that an animal's ability to predict disasters

39 Coren speculated that a dog's ability to predict earthquakes

40 The behaviour of the elephant herds in Wikramanayake's study

A	is released during earthquakes.
B	showed little variation during the tsunami.
C	would have an effect on most dogs.
D	could be linked to their ability to detect movement in the ground.
E	was based on Chinese data-collection methods.
F	could be explained by extra-sensory perception.
G	dates back to before modern civilisation.
H	derived from poorly conducted research.